

Rectified Flow

Based on Q. Liu, *Let Us Flow Together*

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Where we are in the seminar

We have already covered, over the previous sessions:

- Normalizing flows and **flow matching** (Lipman et al.)
- **Stochastic interpolants** (Albergo, Boffi, Vanden-Eijnden)
- Fast sampling
- Discrete diffusion
- Diffusion Schrödinger bridge
- Score matching

Today's pitch. Rectified flow shares most of its core machinery with flow matching and stochastic interpolants. Today we isolate what is specific to Liu's framework.

What is specific to this book?

After removing what overlaps with flow matching / stochastic interpolants:

1. **Reflow**: iterating rectification to straighten trajectories.
2. An **equivariance theorem** showing all affine interpolations are essentially the same object up to a time/space rescaling.
3. A **monotonicity result** on the Euler update giving a direct bridge to L^2 optimal transport.
4. A clean explanation of why **larger diffusion concentrates samples**.
5. A training-free **reward-tilting** formula in the Gaussian case.

Reflow (item 1) is the central contribution; the OT analysis (items 3–4) is the other distinctive thread.

Outline

Part I: A 5-minute recap of rectified flow

Part II: Reflow, the main contribution

Part III: Equivariance of affine interpolations

Part IV: Monotonicity and L^2 optimal transport

Part V: The over-concentration phenomenon

Part VI: Reward tilting in closed form

Conclusion

Part I: A 5-minute recap of rectified flow

The generative modelling problem

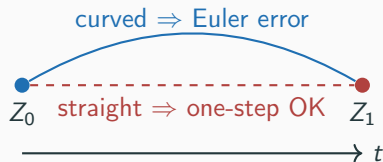
We are given samples from two distributions π_0 (noise) and π_1 (data) on $\mathbb{R}^d$.

We want an ODE

$$dZ_t = v_t(Z_t) dt, \quad Z_0 \sim \pi_0, \quad t \in [0, 1]$$

such that $Z_1 \sim \pi_1$.

There are infinitely many such ODEs. We want **straight** ones: they incur zero discretization error under Euler integration.



Interpolation $\rightarrow$ ODE: the rectification trick

Step 1. Pick a coupling $(X_0, X_1) \sim \pi_0 \times \pi_1$ and build an **interpolation process**:

$$X_t = tX_1 + (1 - t)X_0, \quad \dot{X}_t = X_1 - X_0.$$

$\{X_t\}$ has the right marginals but is **not causal**: you need X_1 to draw X_t .

Step 2. Train a vector field v_t by minimising

$$\min_v \int_0^1 \mathbb{E} \left[\|\dot{X}_t - v_t(X_t)\|^2 \right] dt.$$

The minimizer is

$$v_t^*(x) = \mathbb{E}[\dot{X}_t \mid X_t = x]$$

i.e. the **average slope of all interpolation trajectories passing through x** .

The rewiring picture

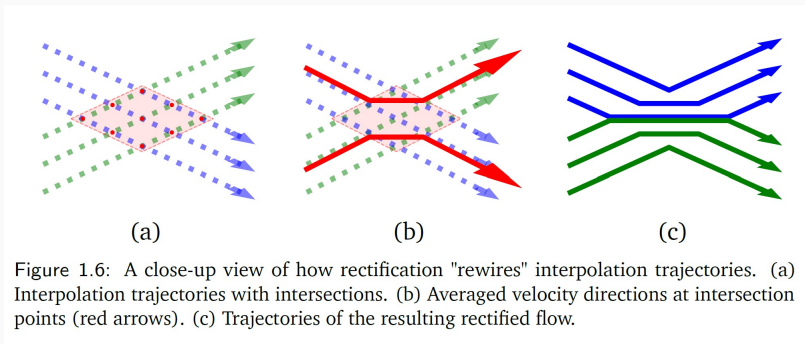


Figure 1.6: A close-up view of how rectification "rewires" interpolation trajectories. (a) Interpolation trajectories with intersections. (b) Averaged velocity directions at intersection points (red arrows). (c) Trajectories of the resulting rectified flow.

At each crossing point x , $v_t^*(x) = \mathbb{E}[\dot{X}_t \mid X_t = x]$ is the *averaged* direction (red arrows in (b)). ODE trajectories cannot cross, so they **follow the average through the crossing region** and exit on their respective sides: the trajectories are **rewired**, not merged.

Two key facts about $\{Z_t\}$ (proof in Appendix A)

Marginal preservation

For $\{Z_t\}$ defined by $dZ_t = v_t^*(Z_t) dt$ with $Z_0 = X_0$:

$$\text{Law}(Z_t) = \text{Law}(X_t), \quad \forall t \in [0, 1].$$

Transport-cost monotonicity

For any convex cost c ,

$$\mathbb{E}[c(Z_1 - Z_0)] \leq \mathbb{E}[c(X_1 - X_0)].$$

The rectified coupling (Z_0, Z_1) is always at least as good as the input coupling, for every convex transport cost simultaneously. This is what motivates iterating.

Part II: Reflow, the main contribution

Why iterate?

After one rectification, $\{Z_t\}$ already has **fewer crossings** than $\{X_t\}$: but it is not straight.

Idea. Feed the new coupling (Z_0, Z_1) back into the algorithm.

$$(Z_0^{k+1}, Z_t^{k+1}, Z_1^{k+1}) = \text{Rectify}(\text{Interp}(Z_0^k, Z_1^k)) \quad k = 0, 1, 2, \dots$$

with $(Z_0^0, Z_1^0) := (X_0, X_1)$.

At each iteration the coupling becomes **more deterministic** (less crossing $\Rightarrow$ each x has a unique tangent).

Quantifying straightness

Definition

The straightness of a process $\{Z_t\}$ is

$$S(\{Z_t\}) = \int_0^1 \mathbb{E}[\|Z_1 - Z_0 - \dot{Z}_t\|^2] dt.$$

$S = 0 \iff \dot{Z}_t = Z_1 - Z_0$ for all $t \iff$ trajectory is a straight line.

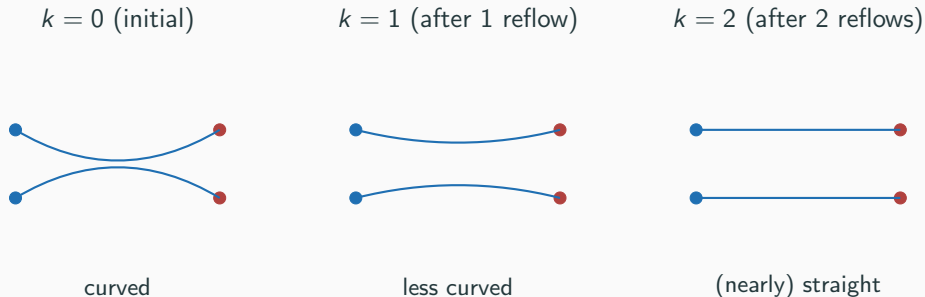
Theorem (Straightening rate)

$$\mathbb{E}_{k \sim \text{Unif}\{1, \dots, K\}}[S(\{Z_t^k\})] = \mathcal{O}(1/K).$$

On average over K reflow iterations, straightness shrinks like $1/K$.

Practical consequence: a sufficiently reflowed model can be sampled in **one Euler step**.

Picture of reflow



Each round straightens the bundle, and the endpoints become **more deterministic** given each other: the coupling (Z_0^k, Z_1^k) converges toward a one-to-one map.

Where reflow breaks down

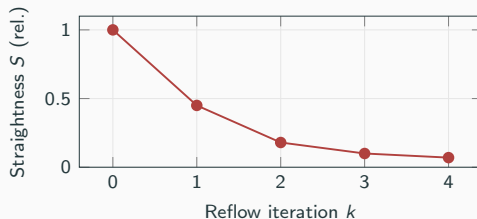
- The rate $\mathcal{O}(1/K)$ is an **average over $k \in \{1, \dots, K\}$** , not a worst-case bound on $S(\{Z_t^K\})$.
- Each round trains on **samples from the previous model**. This resembles **self-distillation**: errors of round k leak into round $k + 1$.
- The bound assumes the rectification is solved *exactly*. In practice we minimise the loss with a finite network on finite data.

New, useful, but *not* a free lunch.

What the empirics actually show

The original rectified-flow paper (Liu, Gong, Liu 2022) and follow-ups report:

- 1-rectified flow alone is competitive with diffusion.
- 2-rectified flow is quite good too.
- Beyond 2–3 reflows, returns diminish sharply: distillation pathology in action.



Part III: Equivariance of affine interpolations

The question

The literature is a zoo of interpolation schemes:

Scheme	α_t	β_t
Straight (rect. flow)	t	$1 - t$
Spherical (DDIM/VP)	$\sin(\pi t/2)$	$\cos(\pi t/2)$
DDPM (exotic)	$\exp(\dots)$	$\sqrt{1 - \alpha_t^2}$
General affine	α_t	β_t

Each has its own "noise schedule" folklore.

Liu's claim: these are all *the same object*, up to a reparametrization of time and a rescaling of the state.

Pointwise transformable processes

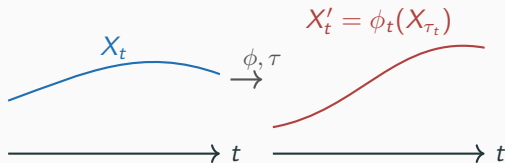
Definition (Liu, Def. 6)

Two processes $\{X_t\}$, $\{X'_t\}$ are **pointwise transformable** if

$$X'_t = \phi_t(X_{\tau_t}), \quad \forall t \in [0, 1],$$

for differentiable maps $\tau : [0, 1] \rightarrow [0, 1]$ and $\phi_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$ (with ϕ_t invertible).

X'_t is just X **at a different time**, viewed through an **invertible spatial deformation**.



The equivariance theorem

Theorem (Liu, Theorem 2 + Corollary 1)

If $\{X_t\}$ and $\{X'_t\}$ are pointwise transformable via (ϕ, τ) with $\tau_0 = 0$, then their rectified flows satisfy the same relation:

$$Z'_t = \phi_t(Z_{\tau_t}), \quad \forall t \in [0, 1].$$

Schematic. The diagram commutes ($\text{Rectify} \circ \text{Transform} = \text{Transform} \circ \text{Rectify}$):



Why is this true? Proof sketch (full in Appendix B)

The rectified trajectories live on the **same "trace"** as the interpolation trajectories. Applying a pointwise deformation ϕ_t **deforms both the interpolation and its rectification by the same map**: rewiring at crossings is insensitive to invertible deformations.

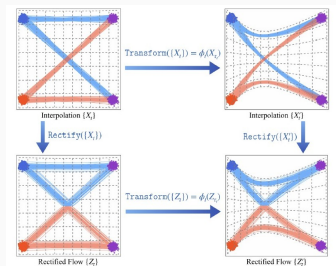


Figure 1: Rectified flow rewires the interpolation trajectories to avoid crossing. When a deformation is applied on the interpolation trajectories, the trajectories of the corresponding rectified flow is deformed in the same way.

Affine interpolations are all pointwise transformable

Lemma (Liu, Lemma 4)

Let $X_t = \alpha_t X_1 + \beta_t X_0$ and $X'_t = \alpha'_t X_1 + \beta'_t X_0$ be two affine interpolations with the same coupling. Then

$$X'_t = \omega_t^{-1} X_{\tau_t}, \quad \text{with} \quad \frac{\alpha_{\tau_t}}{\beta_{\tau_t}} = \frac{\alpha'_t}{\beta'_t}, \quad \omega_t = \frac{\alpha_{\tau_t}}{\alpha'_t} = \frac{\beta_{\tau_t}}{\beta'_t}.$$

So $(\phi_t, \tau_t) = (\omega_t^{-1} \text{Id}, \tau_t)$ is the explicit transform.

Corollary (Same rectified coupling)

$$(Z'_0, Z'_1) = (Z_0, Z_1).$$

All affine interpolations yield the *exact same* rectified endpoints.

Three concrete consequences

1. **Same coupling.** Training on straight vs. spherical vs. DDPM gives the same (Z_0, Z_1) .
2. **Loss equivalence.** Training v_t under interpolation α_t, β_t with weight η_t is equivalent to training under α'_t, β'_t with reweighted loss

$$\eta'_t = \frac{\omega_t^2}{\dot{\tau}_t} \eta_{\tau_t}.$$

3. **DDIM = natural Euler on spherical = vanilla Euler on straight with non-uniform grid.**

In practice: train on the straight interpolation, recover the others by adjusting the time grid and the loss weighting.

Natural Euler sampler

Vanilla Euler: $\hat{z}_{t+\varepsilon} = \hat{z}_t + \varepsilon v_t(\hat{z}_t)$.

For curved interpolations, replace the tangent line with a tangent **interpolation curve**:

Natural Euler update

$$\hat{z}_{t+\varepsilon} = I_{t+\varepsilon}(\hat{x}_{0|t}, \hat{x}_{1|t})$$

where $(\hat{x}_{0|t}, \hat{x}_{1|t})$ are the endpoints of the unique interpolation curve passing through $\hat{z}_t$ with slope $v_t(\hat{z}_t)$ at time t .



DDIM is a natural Euler sampler

Spherical interpolation: $X_t = \sin(\frac{\pi}{2}t)X_1 + \cos(\frac{\pi}{2}t)X_0$.

The natural Euler update simplifies to

$$\hat{z}_{t+\varepsilon} = \cos(\frac{\pi}{2}\varepsilon) \hat{z}_t + \frac{2}{\pi} \sin(\frac{\pi}{2}\varepsilon) v_t(\hat{z}_t).$$

Rewriting in terms of $\hat{x}_{0|t}(\hat{z}_t)$ recovers exactly the DDIM update:

$$\frac{\hat{z}_{t+\varepsilon}}{\alpha_{t+\varepsilon}} = \frac{\hat{z}_t}{\alpha_t} + \left(\frac{\beta_{t+\varepsilon}}{\alpha_{t+\varepsilon}} - \frac{\beta_t}{\alpha_t} \right) \hat{x}_{0|t}(\hat{z}_t).$$

Theorem (Liu, Theorem 5)

For pointwise transformable interpolations, natural-Euler trajectories on grid $\{t'_i\}$ are pointwise transformable to vanilla-Euler trajectories on grid $\{t_i\} = \{\tau(t'_i)\}$.

The equivariance in practice

Mathematically: the equivalence is exact.

In practice::

*"the equivalence holds even when models are approximately optimized using gradient-based optimizers, **provided that the random seeds for initialization and mini-batch sampling are matched exactly.**"*

But seeds are **never** matched across experiments. SGD trajectories diverge under reparametrization. So:

- The space of solutions is the same.
- The optimization landscape is reparametrized.
- **Which minimum you reach in practice** depends on the parametrization.

Part IV: Monotonicity and L^2 optimal transport

Take the canonical setting:

$$X_t = \alpha_t X_1 + \beta_t X_0, \quad X_0 \perp\!\!\!\perp X_1, \quad X_0 \sim \mathcal{N}(0, I), \quad \frac{\dot{\alpha}_t}{\alpha_t} \geq 0 \geq \frac{\dot{\beta}_t}{\beta_t}.$$

The one-step Euler map at time t for step $s - t$:

$$\Phi_{s|t}^{\text{Euler}}(z) = z + (s - t) v_t(z).$$

Question. What can we say about this map?

The monotonicity theorem

Theorem (Liu, Theorem 9)

For $0 \leq s - t \leq -\beta_t / \dot{\beta}_t$, the map $\Phi_{s|t}^{\text{Euler}}$ is *monotone*:

$$(\Phi_{s|t}^{\text{Euler}}(z) - \Phi_{s|t}^{\text{Euler}}(z'))^T (z - z') \geq 0.$$

Moreover, $\Phi_{s|t}^{\text{Euler}} = \nabla f$ for some convex $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

For the straight interpolation: $-\beta_t / \dot{\beta}_t = 1 - t$, so the result holds for *every* Euler step with $0 \leq t \leq s \leq 1$.

Sketch. Use the covariance identity (Lemma 13):

$$\nabla v_t(x) = \frac{\dot{\beta}_t}{\beta_t} I + \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right) \text{Var}(X_0 \mid X_t = x).$$

$\text{Var}(\cdot)$ is PSD, so $\nabla \Phi_{s|t}^{\text{Euler}}(z) = I + (s - t) \nabla v_t(z)$ is PSD under the step size condition. (Full proof in Appendix D.)

Why we care: Brenier's theorem

Brenier (informal)

The L^2 optimal transport map between two measures μ, ν on $\mathbb{R}^d$ is the (unique) **gradient of a convex function** that pushes μ onto ν .

So $\Phi_{s|t}^{\text{Euler}}$ **is** the L^2 optimal transport map between π_t and the distribution of $\hat{Z}_s = \Phi_{s|t}^{\text{Euler}}(Z_t)$.

From monotonicity to optimal transport

Theorem (Liu, Theorem 10: straight interp)

For the rectified flow of $X_t = tX_1 + (1 - t)X_0$:

$$\left| \mathbb{E}[\|Z_1 - Z_0\|^2]^{1/2} - W_2(\pi_0, \pi_1) \right| \leq 2 \mathbb{E}[\|Z_1 - Z_0 - v_0(Z_0)\|^2]^{1/2}.$$

- LHS measures how far (Z_0, Z_1) is from being an L^2 -optimal coupling.
- RHS is the **one-step Euler error at $t = 0$** .
- **If one-step Euler is exact, then (Z_0, Z_1) is the L^2 optimal transport plan.**

This gives a clean operational meaning to "perfectly straight = optimally coupled".

Full ODE map is *not* monotone in general

Remark 23 (Liu). The composition of monotone maps is not necessarily monotone in $d \geq 2$.

The product AB of PSD matrices is PSD **only if A and B commute**.

So the full transport map $\Phi_{1|0}$ (composition of many infinitesimal Euler steps) need not be a gradient of a convex function.

Remark 24. It is only **orientation preserving** ($\det(\nabla\Phi) > 0$).

Take-away. The OT-optimality result is one-step; it does not say the rectified flow solves OT in general.

Part V: The over-concentration phenomenon

Adding noise: SDE samplers

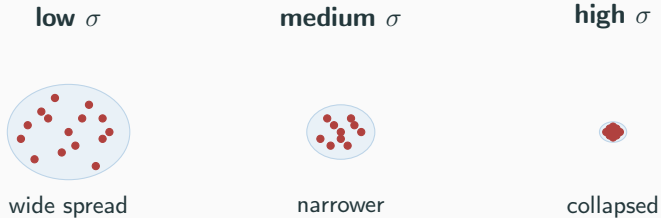
Compound the ODE with the Langevin dynamics of $\rho_t = \text{Law}(Z_t)$:

$$d\tilde{Z}_t = \underbrace{v_t(\tilde{Z}_t) dt}_{\text{rect. flow}} + \underbrace{\sigma_t^2 \nabla \log \rho_t(\tilde{Z}_t) dt + \sqrt{2} \sigma_t dW_t}_{\text{Langevin guardrail on } \rho_t}.$$

Theory: Same marginals as the ODE. The Langevin term is in equilibrium.

Why bother? The Langevin term acts as a **negative feedback**: if $\tilde{Z}_t$ drifts off ρ_t (numerical or model error), it gets pulled back.

The counterintuitive observation



More diffusion produces *more concentrated samples*, not more spread.

This is the opposite of the naive ODE-vs-SDE intuition.

Why?

For affine interpolation with Gaussian X_0 (Tweedie):

$$\nabla \log \rho_t(x) = \frac{1}{\beta_t^2} \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right)^{-1} (v_t(x) - \frac{\dot{\alpha}_t}{\alpha_t} x).$$

As $t \rightarrow 1$, $\beta_t \rightarrow 0$, so the prefactor $1/\beta_t^2 \rightarrow \infty$.

Two regimes:

- *True score* $\nabla \log \rho_t$ stays bounded (the data is finite-variance).
- *Estimated score* $\nabla \log \hat{\rho}_t$ inherits the $1/\beta_t^2$ blow-up because $\hat{v}_t \neq v_t$.

The score points **toward modes**. Overshoot the score $\Rightarrow$ samples collapse onto modes.

Practical impact: cartoonish, over-smoothed images in text-to-image generation.

Part VI: Reward tilting in closed form

The reward-tilting problem

Pretrained v_t targets π_1 . Inference-time goal: sample from

$$\pi_1^r(x) \propto \pi_1(x) r(x)$$

for a reward $r \geq 0$ **without retraining**.

General answer (Liu, Theorem 12):

$$v_t^r(x) = \frac{\mathbb{E}[r(X_1) \dot{X}_t \mid X_t = x]}{\mathbb{E}[r(X_1) \mid X_t = x]}.$$

Generally intractable, it requires a model that knows r .

But in a special case, there is a **closed-form, training-free** solution.

Gaussian reward: closed-form tilting

For $r(x) = \exp(-\eta \|x - x^*\|^2/2)$ (Gaussian reward toward x^*):

Lemma 20

$$\hat{x}_{1|t}^r(x) = \hat{x}_{1|\tilde{t}}(\tilde{x})$$

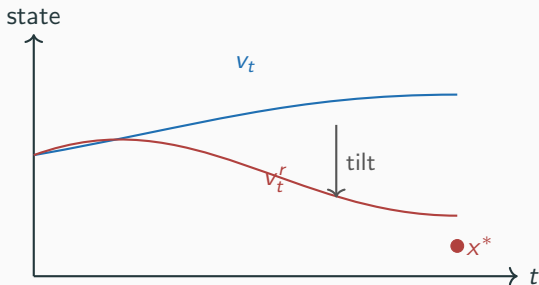
where the **tilted time** $\tilde{t}$ and **tilted position** $\tilde{x}$ solve

$$\frac{\alpha_{\tilde{t}}^2}{\beta_{\tilde{t}}^2} = \frac{\alpha_t^2}{\beta_t^2} + \eta, \quad \tilde{x} = \frac{\beta_{\tilde{t}}^2}{\alpha_{\tilde{t}}} \left(\frac{\alpha_t}{\beta_t^2} x + \eta x^* \right).$$

Inference procedure: given (x, t) , compute $(\tilde{t}, \tilde{x})$, query the original model at $(\tilde{x}, \tilde{t})$, convert back.

A Gaussian tilt is implemented as a **lookup at a different (time, point)** in the same model. No fine-tuning, no extra training.

Tilting picture



x^* **attracts** (for $\eta > 0$) or **repels** (for $\eta < 0$) the sample.

Caveat. Negative η creates a singularity near $t = 0$; mitigated by a threshold in Algorithm 2.

Conclusion

Where rectified flow stands vs. neighbours

	Flow Matching	Stoch. Interpolants	Rectified Flow
ODE-only formulation	yes	yes	yes
Affine interpolations	yes	yes	yes
General coupling	yes	yes	yes
Reflow / iteration	no	no	yes
Monotonicity / one-step OT bound	no	no	yes
Reward tilting (training-free)	no	no	Gaussian case

The three frameworks have **nearly identical training losses**. The distinctive features of Liu's book are reflow and the OT-style analysis.

What to remember:

1. Rectified flow = flow matching + reflow iteration.
2. All affine interpolations are pointwise transformable; the choice is, in theory, just a time/loss reparametrization.
3. One-step Euler = L^2 -OT map (when exact). The full ODE map is *not* OT-optimal in general.
4. Stronger diffusion in SDE samplers *concentrates* samples: score over-estimation, not noise.
5. Gaussian reward tilting has a clean training-free formula.

What to be skeptical of:

- Reflow as a free lunch (it isn't; inherits distillation pathologies).
- Affine-interp. equivalence in practice (SGD breaks the equivariance).

Thank you.

Questions, objections, counter-examples.

Appendix

Appendix overview

- A.** Marginal preservation: full proof.
- B.** Equivariance theorem: full proof.
- C.** Tweedie's formula for affine interpolations.
- D.** Monotonicity of the Euler update: full proof.
- E.** The L^2 -OT bound: full proof.

Appendix A: Marginal preservation

Claim. $\text{Law}(Z_t) = \text{Law}(X_t)$ for all t , where $dZ_t = v_t^*(Z_t) dt$, $Z_0 = X_0$, $v_t^*(x) = \mathbb{E}[\dot{X}_t \mid X_t = x]$.

Proof. For any test function $h \in C_c^1(\mathbb{R}^d)$,

$$\frac{d}{dt} \mathbb{E}[h(X_t)] = \mathbb{E}[\nabla h(X_t)^\top \dot{X}_t] = \mathbb{E}[\nabla h(X_t)^\top \mathbb{E}[\dot{X}_t \mid X_t]] = \mathbb{E}[\nabla h(X_t)^\top v_t^*(X_t)].$$

This gives, with $\pi_t := \text{Law}(X_t)$:

$$\int h d\pi_t - \int h d\pi_0 = \int_0^t \int \nabla h^\top v_s^* d\pi_s ds. \quad (*)$$

By Kurtz (2011) Corollary 1.3, $(*)$ uniquely identifies $\{\pi_t\}$ given π_0 .

$\text{Law}(Z_t)$ satisfies the same equation (with v_t^* as drift) and starts at π_0 , so by uniqueness $\text{Law}(Z_t) = \text{Law}(X_t)$. □

Appendix B: Equivariance, full proof (1/2)

Setup. $X'_t = \phi_t(X_{\tau_t})$ with ϕ_t, τ_t differentiable, ϕ_t invertible, $\tau_0 = 0$.

Step 1: Compute v'_t .

$$\begin{aligned} v'_t(x) &= \mathbb{E}[\dot{X}'_t \mid X'_t = x] \\ &= \mathbb{E}[\partial_t \phi_t(X_{\tau_t}) + \nabla \phi_t(X_{\tau_t})^\top \dot{X}_{\tau_t} \dot{\tau}_t \mid X_{\tau_t} = \phi_t^{-1}(x)] \\ &= \partial_t \phi_t(\phi_t^{-1}(x)) + \nabla \phi_t(\phi_t^{-1}(x))^\top v_{\tau_t}(\phi_t^{-1}(x)) \dot{\tau}_t. \end{aligned}$$

Step 2: Verify $z'_t := \phi_t(z_{\tau_t})$ solves the ODE for v'_t .

$$\begin{aligned} \dot{z}'_t &= \partial_t \phi_t(z_{\tau_t}) + \nabla \phi_t(z_{\tau_t})^\top \dot{z}_{\tau_t} \dot{\tau}_t \\ &= \partial_t \phi_t(z_{\tau_t}) + \nabla \phi_t(z_{\tau_t})^\top v_{\tau_t}(z_{\tau_t}) \dot{\tau}_t \\ &= v'_t(z'_t) \quad \text{by Step 1.} \end{aligned}$$

Appendix B: Equivariance, full proof (2/2)

Step 3: Uniqueness. Conversely, suppose $\dot{z}'_t = v'_t(z'_t)$. Then

$$\frac{d}{dt}(z'_t - \phi_t(z_{\tau_t})) = v'_t(z'_t) - v'_t(\phi_t(z_{\tau_t})).$$

With matching initial conditions $z'_0 = \phi_0(z_{\tau_0})$, uniqueness of the ODE gives

$$z'_t = \phi_t(z_{\tau_t}), \quad \forall t.$$

Step 4: Initial condition. The rectified flow defaults to $Z'_0 = X'_0$. From $\tau_0 = 0$:

$$Z'_0 = \phi_0(Z_{\tau_0}) = \phi_0(Z_0) = \phi_0(X_0) = X'_0. \quad \checkmark$$

Step 5: Same coupling. If $\tau_0 = 0$, $\tau_1 = 1$ and X'_t, X_t share the same coupling, then $\phi_0 = \text{Id}$ and $\phi_1 = \text{Id}$ on the support, so

$$(Z'_0, Z'_1) = (\phi_0(Z_0), \phi_1(Z_1)) = (Z_0, Z_1). \quad \square$$

Appendix C: Tweedie identity for affine interpolations

Setting. $X_t = \alpha_t X_1 + \beta_t X_0$, $X_0 \perp\!\!\!\perp X_1$, $X_0 \sim \mathcal{N}(0, I)$.

Conditional log-density of X_t given $X_1 = x_1$:

$$\log \rho_{X_t|X_1}(x | x_1) = -\frac{1}{2\beta_t^2} \|x - \alpha_t x_1\|^2 - d \log \beta_t + \text{const.}$$

Taking the gradient in x :

$$\nabla \log \rho_{X_t|X_1}(x | x_1) = -\frac{1}{\beta_t^2} (x - \alpha_t x_1).$$

Now marginalize over X_1 (using the score-of-mixture identity

$\nabla \log \rho_X(x) = \mathbb{E}[\nabla \log \rho_{X|Z}(X | Z) | X = x]$):

$$\nabla \log \rho_t(x) = -\frac{1}{\beta_t} \mathbb{E}[X_0 | X_t = x] = -\frac{1}{\beta_t} \hat{x}_{0|t}(x).$$

Combine with the linear identity $v_t = \frac{\dot{\alpha}_t}{\alpha_t} x - \beta_t \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right) \hat{x}_{0|t}$ to express $\nabla \log \rho_t$ via v_t (eq. (4.1) in the book).

Appendix D: Monotonicity, full proof

Lemma (Liu, 13). With $X_0 \sim \mathcal{N}(0, I)$ independent of X_1 :

$$\nabla v_t(x) = \frac{\dot{\beta}_t}{\beta_t} I + \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right) \text{Var}(X_0 \mid X_t = x).$$

Proof of Theorem 9. For the Euler map $\Phi_{s|t}^{\text{Euler}}(z) = z + (s - t)v_t(z)$,

$$\nabla \Phi_{s|t}^{\text{Euler}}(z) = I + (s - t) \nabla v_t(z) = \left(1 + (s - t) \frac{\dot{\beta}_t}{\beta_t} \right) I + (s - t) \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right) \text{Var}(X_0 \mid X_t).$$

$\text{Var}(X_0 \mid X_t) \succeq 0$. Under the assumptions $\dot{\alpha}_t/\alpha_t \geq 0 \geq \dot{\beta}_t/\beta_t$ and $0 \leq s - t \leq -\beta_t/\dot{\beta}_t$:

$$1 + (s - t) \frac{\dot{\beta}_t}{\beta_t} \geq 0, \quad (s - t) \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right) \geq 0,$$

so $\nabla \Phi_{s|t}^{\text{Euler}}(z) \succeq 0$.

A continuous map whose Jacobian is symmetric and PSD everywhere is the gradient of a convex function (Helmholtz + Frobenius). Hence $\Phi_{s|t}^{\text{Euler}} = \nabla f$ for f convex. □

Appendix E: The L^2 -OT bound, full proof

Notation. $\pi_t = \text{Law}(Z_t)$. $\hat{\pi}_{s|t} = \text{Law}(\hat{Z}_{s|t})$ where $\hat{Z}_{s|t} = \Phi_{s|t}^{\text{Euler}}(Z_t)$.

By Appendix D, $\Phi_{s|t}^{\text{Euler}}$ is the gradient of a convex function. By **Brenier's theorem**, it is the L^2 OT map from π_t to $\hat{\pi}_{s|t}$:

$$W_2(\pi_t, \hat{\pi}_{s|t}) = \mathbb{E}[\|\hat{Z}_{s|t} - Z_t\|^2]^{1/2}.$$

Triangle inequality + the bound $W_2(\hat{\pi}_{s|t}, \pi_s) \leq \mathbb{E}[\|\hat{Z}_{s|t} - Z_s\|^2]^{1/2}$:

$$\begin{aligned}\mathbb{E}[\|Z_s - Z_t\|^2]^{1/2} &\leq \mathbb{E}[\|\hat{Z}_{s|t} - Z_t\|^2]^{1/2} + \mathbb{E}[\|\hat{Z}_{s|t} - Z_s\|^2]^{1/2} \\ &= W_2(\pi_t, \hat{\pi}_{s|t}) + \mathbb{E}[\|\hat{Z}_{s|t} - Z_s\|^2]^{1/2} \\ &\leq W_2(\pi_t, \pi_s) + W_2(\hat{\pi}_{s|t}, \pi_s) + \mathbb{E}[\|\hat{Z}_{s|t} - Z_s\|^2]^{1/2} \\ &\leq W_2(\pi_t, \pi_s) + 2 \mathbb{E}[\|\hat{Z}_{s|t} - Z_s\|^2]^{1/2}.\end{aligned}$$

Setting $t = 0, s = 1$ gives the stated bound. □